

Mahbub Sir's Part

Reference Book: Abraham Silberschatz - Operating System Concepts [10th Edition] (2018)

Suggested Playlist: Operating System by Neso Academy

Note: Though Sir basically followed slides (mostly all of it) in class.

Chapter 1 : Introduction

Important Topics:

- I/O Subsystem ★★

Chapter 2 : Operating-System Structures

Important Topics:

- User Goals and System Goals ★★★ [Viva]

Chapter 3 : Processes

Chapter 4 : Threads and Concurrency

Important Topics:

- Resource Sharing ★★★
- Multicore and Multiprocessor Challenges ★★★

Chapter 5 : CPU Scheduling

Important Topics:

- Scheduling Criteria (Definition) ★
- Round-Robin ★★★
- Priority Schedule ★★★

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Note: Sir followed both slides and book, specially during black-out days.

Chapter 6 : Synchronization Tools

- Concurrent Process

Definition: Cooperating Process

6.1 Background

- Producer-Consumer Code
 - Race Condition

6.2 Critical Section Problem

1. Mutual Exclusion
2. Progress
3. Bounded waiting

6.3 Peterson's Solution

Figure- 6.3, 6.4

6.4 Hardware Support for Synchronization

6.4.1 Memory Barriers

6.4.2 Hardware Instructions Figure 6.5

- Atomic Process

Figure- 6.7, 6.8, 6.9

Mutex Locks

- Mutual Exclusion

Figure 6.10

Lock Contention

What is meant by "short duration"?

6.6 Semaphore

6.7 Monitors

6.7.1 Monitor Usage Abstract Data Type - ADT

Function-based Solution -> Monitor

Figure 6.11

6.7.2 Implementing a Monitor using Semaphores

How Monitors are used for reasoning?

6.7.3 Resuming Process within a Monitor

6.8 Liveness

6.8.1 Deadlocks

6.8.2 Priority

Chapter 7 : Synchronization Examples

Problems in Chapter 6 - Race Condition

Three Problems: 1. Bounded-Buffer 2. Readers-Writers 3. Dining Philosophers

7.1.1 The Bounded-Buffer Problem

- unshared data structure

Figure- 7.1, 7.2

7.1.2 The Reader-Writer Problem

- First Reader-Writer Problem
- Second Reader-Writer Problem

Figure- 7.3, 7.4

7.1.3 The Dining Philosophers Problem Figure- 7.6, 7.7

Chapter 8 : Deadlocks

Process: A process is a program in execution.

Thread: A thread is a basic unit of CPU utilization.

8.1 System Model

8.2 Deadlock in Multithreaded Application

8.2.1 Livelock

8.3 Deadlock Characterization

8.3.1 Necessary Conditions

8.3.2 Resource Allocation Graph

8.4 Methods for Handling Deadlocks

8.5 Deadlock Prevention

8.5.1 Mutual Exclusion

8.5.2 Hold and Wait

8.5.3 No Preemption

8.5.4 Circular Wait

8.6 Deadlock Avoidance

- Low Device Utilization
- Reduced System Throughput

8.6.1 Safe State

8.6.2 Resource Allocation Graph Algorithm Figure- 8.9, 8.10

- Claim edge
- Assignment

8.6.3 Banker's Algorithm

- Data Structure
 - Available
 - Max
 - Allocation
 - Need

8.6.3.1 Safety Algorithm

8.6.3.2

8.6.3.3 An Illustrative Example

8.7 Deadlock Detection

8.7.1 Single Instances of Each Resource Type Figure 8.11

8.7.3 Deletion-Algorithm Usage

Self Reading

8.8 Recovery from Deadlock

8.8.1 Process and Tread Termination

- Abort all deadlock processes

From Page-342, ★★

Many factors may affect which process is chosen, including:

1. What the priority of the process is
2. How long the process has computed and how much longer the process will compute before completing its designated task
3. How many and what types of resources the process has used (for example, whether the resources are simple to preempt)
4. How many more resources the process needs in order to complete
5. How many processes will need to be terminated

8.8.2 Resource Preemptioin

Chapter 9 : Main Memory

Figure 9.1

9.1 Address Binding

- Compile Time
- Execution Time

9.1.3 Logical Versus Physical Address Space Figure 9.4

9.3 Paging

9.3.1 Basic Method Figure 9.8